

PROBLEM STATEMENT #38 | Retail, E-Commerce & Finance

Dropshipping Inventory & Order Orchestrator

1. Problem Context & Overview

An automated e-commerce backend that ingests storefront webhooks, verifies third-party supplier stock availability in real-time, splits multi-vendor orders, and tracks tracking numbers.

Target Stakeholders / Actors: *E-Store Owner, Supplier Partner*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.

Sample Acceptance Criteria: Pass: Supplier PO triggered with customer delivery address; Fail: Order placed for out-of-stock supplier SKU.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storm.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.